to get only the product ID, the title, a short description and list price of a given product. Let us see how we can query to get the other details we're interested in.

To demonstrate GraphQL I have used the GUI based plugin GraphiQL. For consuming from other applications, we can configure the endpoint in *application.properties*.

```
graphql.servlet.mapping=/graphql
graphql.servlet.enabled=true
graphql.servlet.corsEnabled=true
```

Now we can make a call to the above endpoint by passing a URL encoded query parameter as shown in Figure 7. You can learn more about queries and mutations at *https://graphql.org/learn/queries/*. END

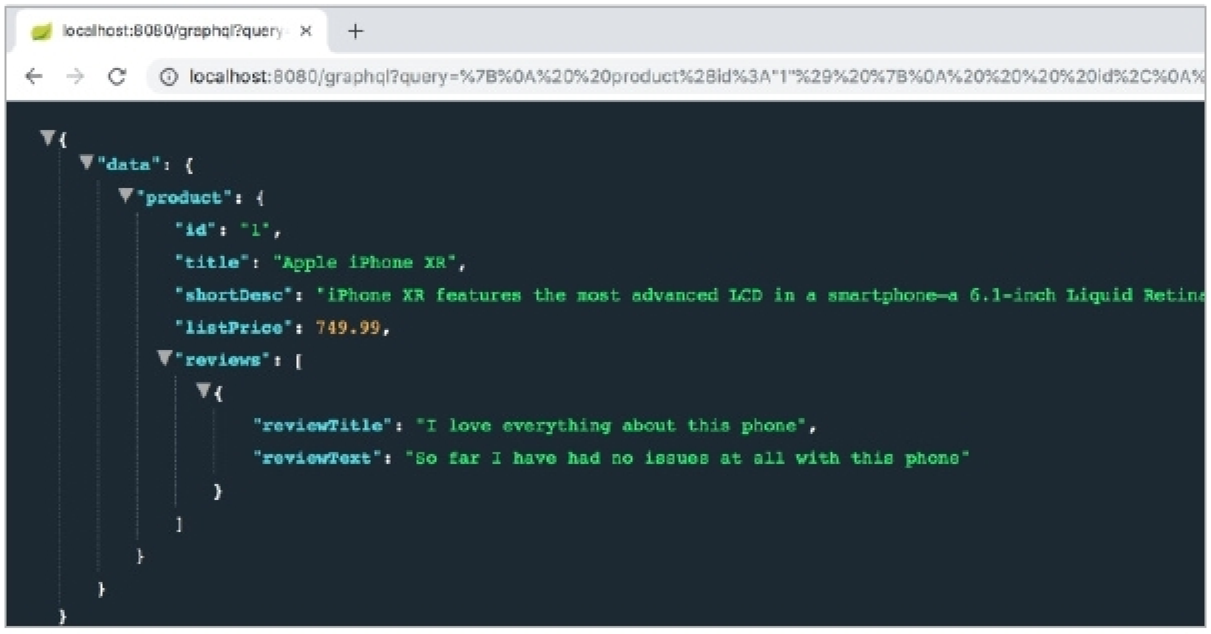

Figure 7: GraphQL_query

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After loading the saved model and invoking the testing data set, the re-evaluation of the model is done specifically for the supplied test set. It is used so that the supplied testing data set can be assigned the predicted classes on the same mathematical functions and model as used in the earlier classification model.

The predicted classes can be viewed in the right side panel of Weka after re-evaluation of the model. As in Figure 12, there are multiple attributes including the instance number, actual and predicted. In the predicted attribute, the unknown class (represented as'?') is assigned with the determined class on the same algorithm of machine learning. For example, in the first instance, the predicted class is 1, which was unknown in the test data set. In a similar way, the other values can be predicted.

Using WekaDeeplearning4j for deep learning in Weka

WekaDeeplearning4j *(https://deeplearning.cms.waikato.ac.nz/)* is the dedicated package for the implementation of deep learning in different applications. This library has been released so that the features and accuracy of deep learning can be used with the data analytics and predictive mining based applications. The power of Java programming is associated at the back-end of Weka with deep learning modules.

The following are the key features and layers integrated in Weka with deep learning:

- Convolution layer
- Dense layer
- Sub-sampling layer
- Batch normalisation
- Long short term memory (LSTM)
- Global pooling layer
- Output layer

Scope for research and development

The algorithms of machine learning, deep learning, data science and knowledge discovery are closely associated with scientific and engineering applications. They can be used for the development of new algorithms and for solving the engineering optimisation problems in the social as well as scientific domains. These algorithms have custom functions that can be updated as per the requirements of dynamic data sets to achieve a higher degree of accuracy and performance with related degrees of effectiveness. END

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